



Simulation of Model_Mania_2008

Date: Hidden
Designer: Solidworks
Study name: SimulationXpress Study
Analysis type: Static

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Description

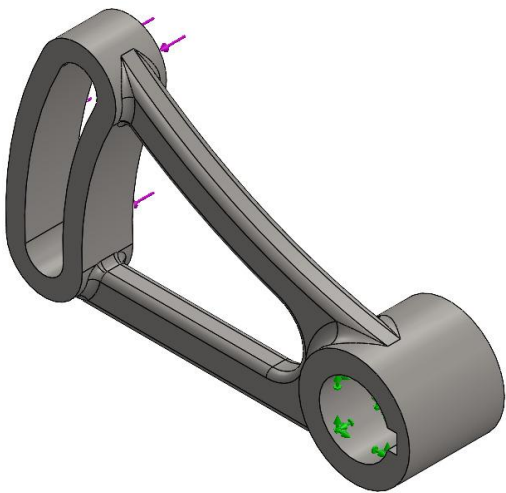
Model_Mania_2008-SimulationXpress Study-1

Fixture and Forces were defined as asked for in Phase 2 Instructions.



Assumptions

Model Information



Model name: Model_Mania_2008
Current Configuration: Phase 2

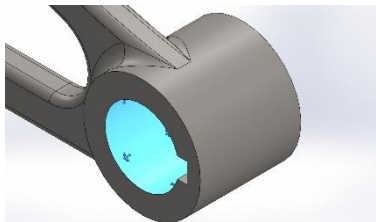
Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Fillet4 	Solid Body	Mass:0.371553 kg Volume:4.7032e-05 m^3 Density:7,900.01 kg/m^3 Weight:3.64122 N	Hidden

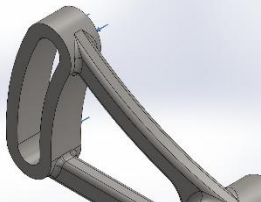


Material Properties

Model Reference	Properties	Components
	Name: AISI 1020 Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 3.51571e+08 N/m ² Tensile strength: 4.20507e+08 N/m ²	SolidBody 1(Fillet4)(Model_Mania_2008)

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry

Load name	Load Image	Load Details
Force-1		Entities: 1 face(s) Type: Apply normal force Value: 400 N



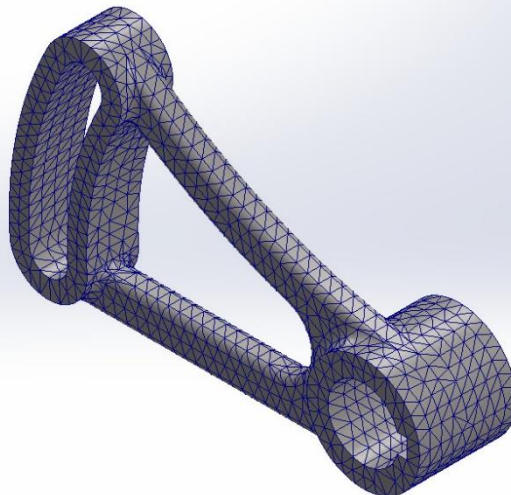
Mesh information

Mesh type	Solid Mesh
Mesher Used:	Standard mesh
Automatic Transition:	Off
Include Mesh Auto Loops:	Off
Jacobian points for High quality mesh	16 Points
Element Size	3.61085 mm
Tolerance	0.180542 mm
Mesh Quality	High

Mesh information - Details

Total Nodes	15364
Total Elements	8963
Maximum Aspect Ratio	15.288
% of elements with Aspect Ratio < 3	97.9
Percentage of elements with Aspect Ratio > 10	0.0669
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:07
Computer name:	N1KUL

Model name: Model_Mania_2008
Study name: SimulationXpress Study(-Phase 2-)
Mesh type: Solid Mesh



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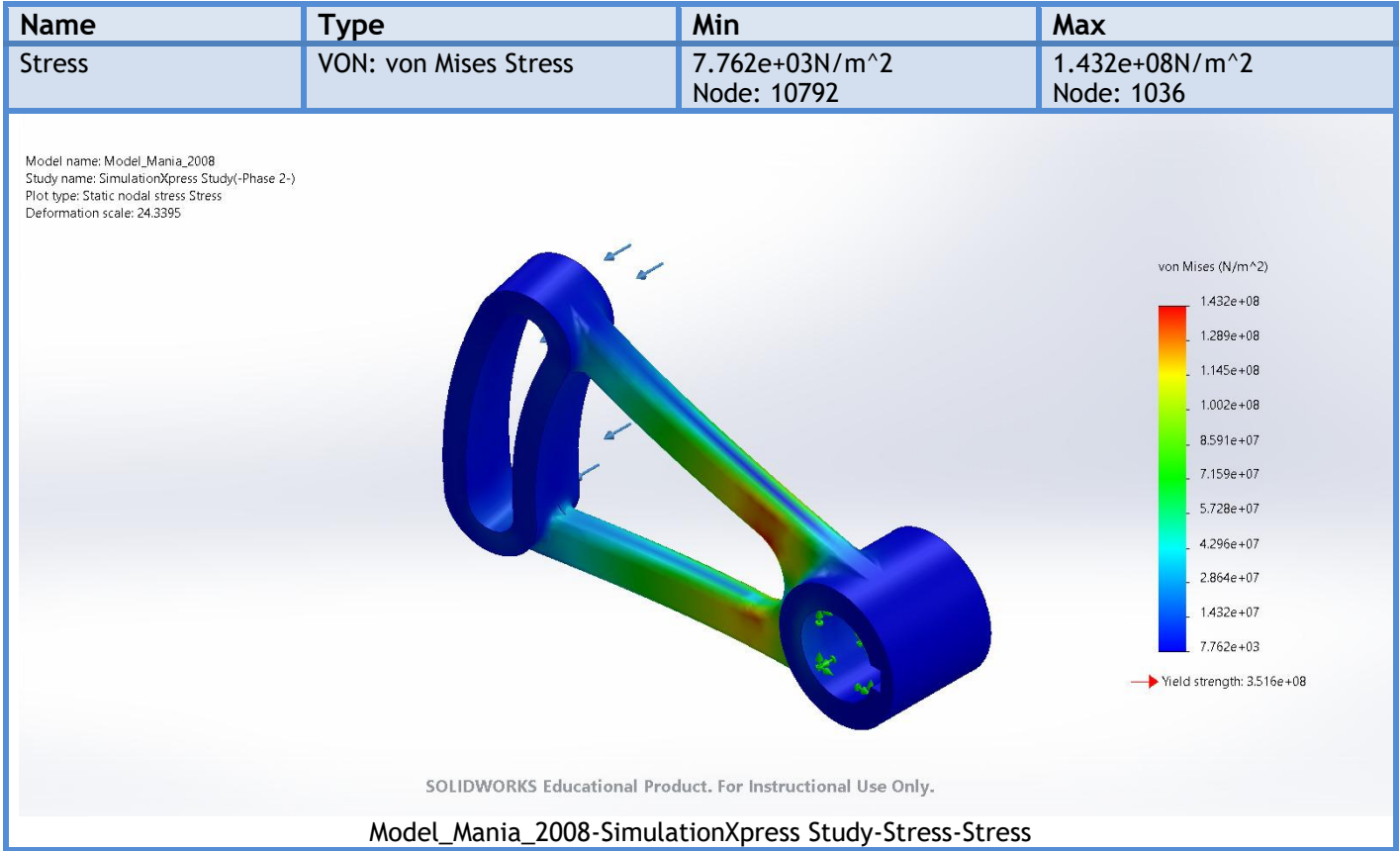
SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of Model_Mania_2008

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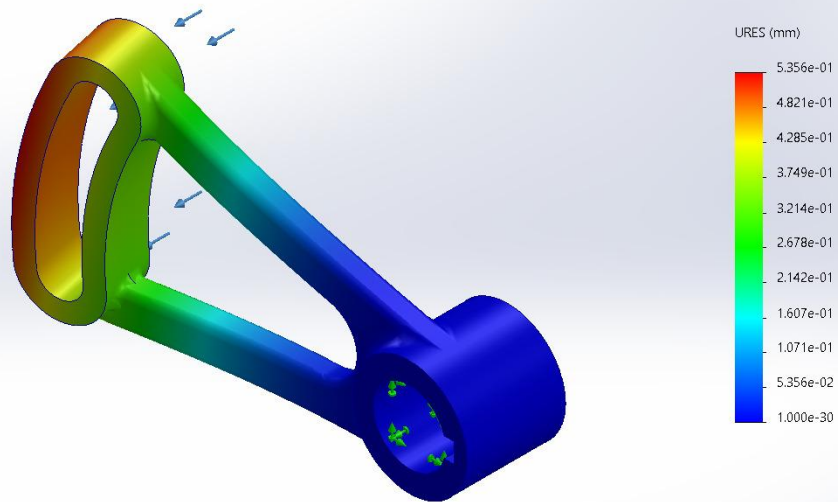
Study Results



Name	Type	Min	Max
Displacement	URES: Resultant Displacement	0.000e+00mm Node: 175	5.356e-01mm Node: 1330



Model name: Model_Mania_2008
 Study name: SimulationXpress Study(-Phase 2-)
 Plot type: Static displacement Displacement
 Deformation scale: 24.3395

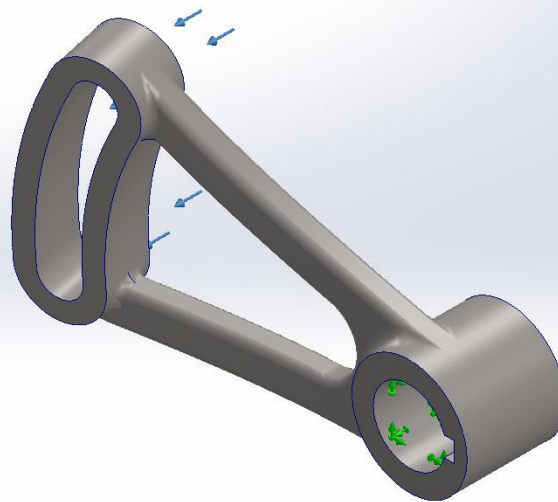


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Model_Mania_2008-SimulationXpress Study-Displacement-Displacement

Name	Type
Deformation	Deformed shape

Model name: Model_Mania_2008
 Study name: SimulationXpress Study(-Phase 2-)
 Plot type: Deformed shape Deformation
 Deformation scale: 24.3395



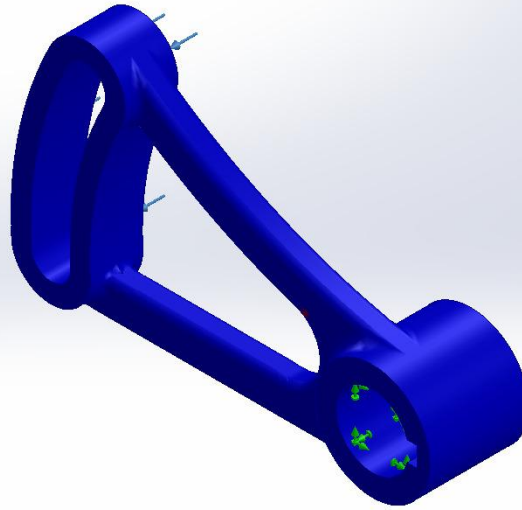
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Model_Mania_2008-SimulationXpress Study-Displacement-Deformation

Name	Type	Min	Max
Factor of Safety	Max von Mises Stress	2.455e+00	4.530e+04



Model name: Model_Mania_2008
Study name: SimulationXpress Study(-Phase 2-)
Plot type: Factor of Safety Factor of Safety
Criterion: Max von Mises Stress
Red < FOS = 2.5 < Blue



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Model_Mania_2008-SimulationXpress Study-Factor of Safety-Factor of Safety

